

Assignment 4

sed

Introduction to UNIX Lab (NCSC102)

Instructions:

- Once completed, show your code to the respective TAs.
 - You may ask the TAs for help if you're encountering syntax errors or having trouble understanding the questions.
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Setup

Create a working directory and populate it with the following files:

```
1 mkdir -p sed_lab && cd sed_lab
2
3 cat > inventory.csv <<'CSV'
4 sku,name,qty,price,location
5 A12-7,Desk Lamp,7,39.99,Aisle-4
6 B04-2,Office Chair,0,129.00,Aisle-1
7 Z99-9,Standing Desk,3,499.00,Aisle-3
8 M31-5,HDMI Cable,14,9.50,Bin-7
9 CSV
10
11 cat > students.txt <<'TXT'
12 # cohort: spring-2025
13 Ava Li|CS|3.8
14 Noah Patel|EE|3.1
15 Maya Gomez|ME|3.6
16 Leo O'Neill|CS|2.9
17 Zara Chan|CE|3.4
18 TXT
19
20 cat > weblog.txt <<'LOG'
21 2025-08-21 10:02:11 INFO /health 200 ok
22 2025-08-21 10:03:44 warn /api/v2/users 200 slow=420ms
23 2025-08-21 10:04:09 error /api/v2/users 500 trace=af12
24 2025-08-21 10:05:33 INFO /login 302 loc=/home
25 2025-08-21 10:06:10 error /payments 503 trace=beef
26 LOG
27
28 cat > names.txt <<'N'
29 Alice Smith
30 Bob Stone
31 Chlo   de la Cruz
32 N
33
34 cat > system.conf <<'CONF'
35 # base config
36 mode=debug
37 cache=true
38 timeout=2500
```

```

39 # feature flags
40 feat.search=on
41 feat.payments=off
42 CONF
43
44 cat > orders.tsv <<'TSV'
45 #id user item amount status
46 101 u01 desk-lamp 39.99 paid
47 102 u91 chair 129.00 refund-requested
48 103 u77 standing-desk 499.00 pending
49 104 u01 hdmi-cable 9.50 paid
50 TSV
51
52 cat > notes.md <<'MD'
53 ## Release
54 - tag: v1.4
55 - date: 2025-08-22
56
57 ## Changelog
58 - fix: payments retry
59 - add: bulk user import
60 MD

```

USE

- Use GNU **sed** for all commands.
- Focus on **sed** usage rather than regex basics (covered previously).
- Unless instructed, do *not* edit files in place.
- Collect all solutions into a script file **solutions.sh**.

Exercises

A. Addressing & Printing

1. Print the first 3 data lines of **inventory.csv** (skip header).
2. Print every 2nd student from **students.txt**.
3. Display only non-comment lines in **orders.tsv**.
4. From **notes.md**, print all lines from section **## Changelog** to the end.

B. Editing Commands

1. In **weblog.txt**, uppercase the word “error” and print only modified lines.
2. In **students.txt**, insert the line “- probation threshold: GPA < 3.0 -” before any GPA < 3.0.
3. In **system.conf**, append “# audited” after lines setting **feat.***.
4. In **inventory.csv**, replace any row with **qty=0** by **OUT-OF-STOCK**.
5. Transliterate the *item* column in **orders.tsv** from lowercase to uppercase.

C. Substitute Flags & ERE

1. In **weblog.txt**, replace only the 2nd occurrence of “/” with “ → ”.
2. Using **-E**, convert amounts (e.g., **129.00**) into “**USD:<amount>**” in **orders.tsv**.
3. In **inventory.csv**, replace **Aisle-<n>** with **A<n>** and write modified lines into **aisle.edits**.

D. Multi-file Stream vs Separate Mode (10 min)

1. Create files **part1.txt** and **part2.txt**.
2. Run:

```
1 sed -n '1p' part1.txt part2.txt
2 sed -n -s '1p' part1.txt part2.txt
```

3. Compare outputs and explain the role of the **-s** option.

E. Joining & Flow Control

1. Join any “error” line with the next line, replacing the newline with “– CONTEXT:”.
2. Demonstrate **P** and **D** on **notes.md** to iteratively print and drop lines.
3. Print the first two non-comment lines of **orders.tsv**, then quit.

F. Hold & Pattern Space

1. Swap every two students in **students.txt** using **h;n;p;g;p**.
2. Deduplicate adjacent duplicates in **paths.txt** by saving and comparing with hold space.
3. Append “# generated: ok” after the last line of **system.conf**.

G. File I/O

1. In **inventory.csv**, write products with price ≥ 100 to **premium.csv** while still printing output.
2. In **notes.md**, after the header “## Release”, read and insert contents of **system.conf**.

H. Backreferences & Formatting

1. In **names.txt**, reformat entries as “Last, First”:
 - Collapse multiple spaces.
 - Capture first and last tokens with ERE.
 - For names with more than two tokens, treat the final token as last name.